

EDUTECH SOLUTION

REPORT GENERATION & DATA STORYTELLING

*Comprehensive A/B Testing Performance Analysis &
Strategic Recommendations for Digital Marketing
Campaigns*

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1. EXECUTIVE SUMMARY

1.1 Problem Overview & Key Constraints

In contemporary direct-to-consumer (D2C) ecosystems, electronic commerce architectures, and digital media landscapes, enterprise capital allocation within marketing optimization programs is frequently subjected to high volatility. Organizations regularly implement programmatic changes to their creative assets, targeting parameters, or algorithmic bidding models to reduce acquisition friction and capture incremental market share. However, introducing unverified changes can undermine baseline conversions, inflate Customer Acquisition Costs (CAC), and compress profit margins.

This business report examines empirical data collected from an industrial digital marketing campaign run by **Edutech Solution**. The marketing division designed an experiment consisting of two distinct strategic treatments: the **Control Campaign**, representing the baseline configuration (employing traditional bidding and standard semantic targeting models), and the **Test Campaign**, introducing a revised algorithmic paradigm designed to scale consumer interaction. The overarching challenge was to determine if the newer, higher-cost framework delivered an increase in final macro-conversions that justified the added financial investment.

1.2 Summary of Statistical Outcomes

The data extraction pipeline yielded 60 operational logs spanning August 2019. Quality assurance protocols identified a single null record in the Control dataset, which was removed to avoid bias, resulting in a balanced matrix of 29 Control group observations and 30 Test group observations. A standard descriptive analysis revealed a notable structural shift across the conversion funnel. The Test Campaign expanded mid-funnel interaction, increasing average website clicks by **13.37%** (moving from a baseline of 5,320.79 to 6,032.33 clicks) and increasing user-initiated searches by **8.89%** (climbing from 2,221.31 to 2,418.97 searches).

Crucially, macro-conversions (final completed purchases) remained virtually unchanged between the two frameworks. The Control group generated a mean purchase value of **522.79 units**, while the Test group generated a mean purchase value of **521.23 units**—a marginal decline of 0.30%. To determine if this variation was structural or merely a product of stochastic noise, an Independent Two-Sample T-Test was executed under an alpha benchmark of 0.05. The statistical computations generated a **T-statistic of 0.0302** and an identical **P-value of 0.9760**. Because the P-value vastly exceeds the alpha threshold, we **fail to reject the Null Hypothesis**.

1.3 Primary Strategic Interventions

From a business standpoint, the experiment reveals a clear operational inefficiency: the Test Campaign required **11.24% more financial capital** (\$2,563.07 vs. \$2,304.07) while failing to yield any incremental revenue. This dynamically compressed the Return on Ad Spend (ROAS) from **0.2269** down to **0.2034**, and dragged overall e-commerce conversion efficiency down from **9.83%** to **8.64%**. Consequently, executive management is advised to halt the full-scale deployment of the Test Campaign infrastructure in its current form.

2. INTRODUCTION & STRATEGIC FRAMEWORK

2.1 Context of the Study

Enterprise marketing analytics has evolved beyond basic descriptive calculations into a prescriptive, multi-tiered optimization discipline. In an era marked by shifting user privacy policies, cookieless attribution challenges, and rising digital media pricing, organizations must evaluate ad spend using empirical, evidence-based methods. Marketing frameworks can no longer rely on intuition or uncalibrated optimizations; every dollar deployed must be accounted for using rigorous mathematical models.

2.2 Core Project Objectives

- **Quantitative Validation:** To design, execute, and evaluate an industry-standard quantitative comparison model that tests if the newly formulated Test Campaign delivers a mathematically verifiable lift in user purchase volume relative to the established historical Control baseline.
- **Friction Identification:** To isolate and map specific customer journey drop-offs across the conversion funnel—from initial exposure (impressions) and interest (clicks and searches) to intent (content views and cart adds) and action (final transactions).
- **Capital Efficiency Analysis:** To measure macro-economic metrics like Return on Ad Spend (ROAS), Cost-Per-Click (CPC), and overall conversion efficiency, ensuring that the company's media spend supports a sustainable long-term profit profile.
- **Strategic Reporting:** To synthesize complex data rows, data validation audits, and mathematical distributions into an integrated business intelligence narrative, providing senior leadership with clear, actionable guidance on future strategy.

2.3 Business Problem Statement

The central issue facing the executive leadership team at Edutech Solution is a misalignment between top-funnel ad spend and bottom-funnel revenue generation. Preliminary media metrics show that while the marketing team successfully increased ad spend to drive top-funnel interest, final purchase counts have reached a structural plateau. The Test Campaign incorporates an aggressive automated bidding algorithm designed to capture premium ad placements on high-traffic networks. This model increased average campaign costs by **11.24%**, but it remains unclear if this increased spend translates into a better bottom line.

2.4 About A/B Testing

A/B testing, or split-testing, is a data-driven framework rooted in statistical hypothesis testing. It presents two variants of a specific variable—the baseline Control variant (A) and the modified Test

variant (B)—to randomized user segments to measure behavioral differences. Mathematically, this methodology relies on sampling theory to infer broader population parameters from localized experimental samples. By controlling for external confounding variables, any observable structural change in conversion values can be confidently attributed to the modified campaign elements.

2.5 Importance of Data Storytelling

Data analytics is only as effective as the business decisions it inspires. Raw databases, p-values, and correlation coefficients can be challenging for non-technical stakeholders to interpret. Data storytelling bridges this gap by turning quantitative data into a clear business narrative that explains what happened, why it happened, what it means for the company, and how management should respond. This report uses data storytelling to translate complex metrics into strategic insights.

3. DATASET DESCRIPTION & CLEANING

3.1 Dataset Overview

The data tracking pipeline generated two independent, structured files stored within the company's secure repository: `control_group.csv` and `test_group.csv`. These records log performance data across 30 consecutive operating cycles in August 2019.

- **Number of Rows:** 60 Initial rows (29 Control, 30 Test post-cleaning)
- **Number of Columns:** 11 Operational features

3.2 Features Description & Data Types

Feature Name	Data Type	Description
Campaign Name	String / Object	Categorical indicator identifying the underlying operational treatment group.
Date	String (DD.MM.YYYY)	Temporal stamp mapping the logging interval.
Spend [USD]	Integer (int64)	Total daily financial capital deployed.
# of Impressions	Float (float64)	Total ad views. Raw exposure layer of the conversion funnel.
Reach	Float (float64)	Count of unique users exposed to the ad asset.
# of Website Clicks	Float (float64)	Volumetric count of user clicks redirecting to the target lander.
# of Searches	Float (float64)	Count of user queries performed on the lander.
# of View Content	Float (float64)	Detailed product page views. Active engagement phase.
# of Add to Cart	Float (float64)	Count of high-intent actions where products are added to the queue.
# of Purchase	Float (float64)	Count of completed checkout payments. Primary KPI metric.

3.3 Data Cleaning & Preprocessing

Prior to execution of exploratory transformations or statistical parsing, data records underwent a strict quality assurance audit. Programmatic parsing using null-location functions revealed a critical integrity issue on August 5, 2019, within the Control group tracking array. On this operating day, every single downstream performance metric failed to record, generating NaN fields across the row. Because this missing data was isolated to a single day, dropping the row entirely was the safest approach. This preserved strict data integrity and left the model with 59 highly reliable rows for final analysis. Duplicate checking returned exactly 0 records.

4. EXPLORATORY DATA ANALYSIS (EDA)

Traffic Performance Metric

4.1 Volumetric Analysis & Group Profiles

The aggregated metric breakdown across both operational campaign treatment frameworks highlights significant variance throughout the conversion stages:

Group Profile	Spend [USD]	Impressions	Website Clicks	Searches	Add to Cart	Purchase
Control Campaign	\$2,304.07	109,559.76	5,320.79	2,221.31	1,300.00	522.79
Test Campaign	\$2,563.07	74,584.80	6,032.33	2,418.97	881.53	521.23

Insert Figure 1: Average Funnel Metric Distribution Group Comparison

Objective: Visualize step-down counts between Control and Test frameworks to isolate operational drop-off stages.

Observation & Interpretation: The Test group effectively expanded top-funnel interaction, increasing website clicks by 13.37% and searches by 8.89%. However, downstream engagement collapsed sharply, with Add-to-Cart metrics plunging by 32.19% from 1,300.00 to 881.53. This establishes a structural friction point at the conversion stage.

Insert Figure 2: Correlation Heatmap Matrix (Pearson r Coefficients)

Objective: Examine linear relationships and dependencies among all numeric tracking parameters.

Observation & Interpretation: In the Control group, strong positive correlations exist uniformly across the user journey. In the Test group, the linear link between top-funnel clicks and final

completed transactions breaks down significantly, confirming that the expanded click volume failed to drive operational efficiency downstream.

5. STATISTICAL INFERENCE & HYPOTHESIS TESTING

5.1 Mathematical Hypothesis Formulation

To verify if the experimental outcomes were driven by random variation or a true behavioral shift, the macro-purchase metric served as the primary benchmark variable. The hypotheses were defined as:

Null Hypothesis (H_0): $\mu_{Control} = \mu_{Test}$

Alternative Hypothesis (H_1): $\mu_{Control} \neq \mu_{Test}$

5.2 Independent Two-Sample T-Test Calculations

An Independent Two-Sample Welch's T-Test was executed to evaluate the performance difference between the two campaign models without assuming equal population variances.

Statistical Metric Array	Computed Operational Value
Mean Purchase Metric - Control Group	522.79 units
Mean Purchase Metric - Test Group	521.23 units
Calculated T-Statistic	0.0302
Computed Probability Value (P-value)	0.9760
Configured Significance Alpha Threshold (α)	0.05

5.3 Core Decision Boundary Verification

"Because the calculated P-value of 0.9760 vastly exceeds the pre-established alpha threshold of 0.05, we fail to reject the Null Hypothesis. There is no statistically significant difference between the final purchase volumes generated by the Control Campaign and the Test Campaign."

6. KEY FINDINGS & BUSINESS INSIGHTS

6.1 Top 10 Detailed Analytical Findings

- 1. Capital Spend Inefficiency:** The Test Campaign increased overall spending by 11.24% (\$2,563.07 vs. \$2,304.07), expanding financial risk without delivering any incremental business value.
- 2. Top-Funnel Growth Success:** The Test Campaign successfully expanded top-funnel engagement, driving a 13.37% increase in website traffic and proving the automated bidding system was highly effective at winning competitive placements.
- 3. Active Consideration Growth:** Early-stage user intent grew alongside traffic, with average search volume increasing by 8.89% in the Test group.
- 4. Deep-Funnel Engagement Breakdowns:** This initial momentum stalled as users moved deeper into the site, with average product page views dropping by 4.41% in the Test group.
- 5. Severe Mid-Funnel Conversion Plunge:** The user journey faced a critical bottleneck at the conversion stage, where the Test group's average Add-to-Cart metrics plummeted by 32.19%.
- 6. Stagnant Bottom-Funnel Revenue Outcomes:** This funnel breakdown directly impacted the bottom line, with average purchases in the Test group slipping by a marginal 0.30% (\$521.23 vs. \$522.79).
- 7. Loss of Conversion Efficiency:** Total conversion efficiency (purchases divided by clicks) dropped significantly, falling from 9.83% in the Control group to 8.64% in the Test group.
- 8. Compressed Return on Ad Spend:** This combination of higher spend and flat sales compressed the Return on Ad Spend (ROAS) by 10.36%, dragging it down from 0.2269 in the Control group to 0.2034 in the Test group.
- 9. Funnel Alignment Decoupling:** Correlation testing revealed that the Test Campaign decoupled top-funnel traffic from bottom-funnel revenue, breaking the linear relationship that typically drives conversions.
- 10. Definitive Hypothesis Outfall:** The Welch T-test generated a high P-value of 0.9760, mathematically proving that the Test Campaign failed to deliver a statistically valid performance lift.

6.2 Business Interpretation & Management Roadmap

The diagnostic pattern rules out a lack of audience interest. Instead, it highlights an underlying product line or technical issue, such as price anomalies, uncompetitive bundles, unexpected shipping fees, or slow page load times. The Test Campaign successfully brought motivated buyers to the store, but the digital storefront failed to close the deal, causing the extra ad spend to go to waste.

7. ACTIONABLE RECOMMENDATIONS & CONCLUSION

7.1 Actionable Recommendations (Tactical Roadmap)

- **Halt Test Bidding Frameworks:** Executive leadership should immediately stop scaling the Test Campaign's automated bidding algorithm. This will protect profit margins from being eroded by high-cost, inefficient ad inventory.
- **Return to Control Allocations:** Shift core media budgets back to the stable Control Campaign framework. This model maintains a reliable ROAS baseline (\$0.2269) and keeps Customer Acquisition Costs (CAC) predictable.
- **Introduce Cap Constraints:** If automated bidding models are tested in the future, apply strict bid caps to prevent algorithms from over-bidding and driving up overall conversion costs.
- **Audit Landing Page Technical Performance:** Conduct a thorough user-experience and technical speed audit on primary product pages. Focus on optimizing mobile responsiveness, script loads, and core web vitals to ensure pages load instantly and reduce bounce rates.
- **Address Cart Inefficiencies and Hidden Costs:** Review the checkout flow to eliminate potential purchase triggers like unexpected delivery fees or mandatory account creation screens. Simplify the user interface down to a streamlined, single-click checkout option.
- **Resolve Inventory Pricing Discrepancies:** Ensure that dynamic search landing pages match user expectations. Cross-reference ad promotional offers with actual on-site pricing to eliminate confusing price mismatches that drive users away.
- **Deploy Intent-Driven Remarketing Loops:** Leverage the large influx of top-funnel traffic driven by the Test Campaign. Build segment lists of users who searched the site but didn't convert, and re-engage them with targeted, lower-cost remarketing ads offering limited-time discounts or bundle deals.
- **Introduce Smart Upsell and Cross-Sell Bundles:** To combat flat conversion volumes, focus on increasing Average Order Value (AOV). Deploy automated product recommendation modules on cart pages, matching user selections with relevant add-ons or premium software extensions.

7.2 Conclusion & Future Scope

The final Welch Two-Sample T-test generated a high P-value of 0.9760, confirming that there is no statistically significant difference between the performance of the two campaigns. Future scopes include incorporating Multi-Touch Attribution Integration, conducting granular behavioral segmentation, and expanding algorithmic models to predict customer lifetime values using real-time machine learning engines.

8. REFERENCES & TECHNICAL SOURCES

1. Notebook Reference Source: `Task13.ipynb` — Comprehensive workspace execution path containing data validation engines and Welch Two-Sample T-test models.
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4. Statistical Foundations Framework: Data Science & Quantitative Statistical Analysis Manual (2024). Chapter 8: Evaluation Boundaries for Continuous Two-Sample Independent T-Tests.